Huawei Access Node BTM-ACC-23 Hardware/Environmental Fault Resolution SOP (ALARM_9914, Minor Severity)

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9914

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for a hardware/environmental fault on the Huawei Access Node BTM-ACC-23. The ALARM_9914 code indicates an ASIC memory parity error, line card crash, or power supply redundancy loss. The BTM-ACC-23 node, located at the BTM Layout Aggregation site, operates within the Huawei Access network, utilizing the Access role. The failure domain is characterized by the potential for hardware or environmental faults to impact the node's performance and availability. The underlying protocol mechanics involve the Line Card's (LC) Central Processing Unit (CPU) and Memory, which are responsible for processing and storing traffic forwarding information. The ASIC (Application-Specific Integrated Circuit) memory parity error is a critical issue that can lead to data corruption, packet loss, or even node crashes. The power supply redundancy loss can result in a single point of failure, compromising the node's high availability. This SOP aims to identify and troubleshoot the root cause of the issue, followed by resolution steps to restore normal node operation.

Pre-requisite Checks

- Check: Verify that the redundant link status is active, Verification Command: display link info | include BTM-ACC-23, Description: Ensure the redundant link is operational before proceeding with the diagnostic and resolution steps.

- Check: Validate the power supply redundancy status, Verification Command: display power supply info | include BTM-ACC-23, Description: Confirm that the power supply redundancy is operational to prevent a single point of failure.

- Check: Dispatch remote hands to the BTM Layout Aggregation site, Description: Ensure on-site personnel are available to assist with physical checks and access to the node.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display linecard info | include BTM-ACC-23, Expected Output: Line Card Information: Card Type: Access Line Card Status: Down, Description: This command isolates the line card status, which will indicate if the line card is down due to a hardware or software issue.

Command: display power supply info | include BTM-ACC-23, Expected Output: Power Supply Information: Power Supply Status: Normal Redundancy: Active, Description: This command verifies the power supply status and redundancy, ensuring that the power supply redundancy is operational.

Command: display asic info | include BTM-ACC-23, Expected Output: ASIC Information: ASIC Type: Access ASIC Status: Down, Description: This command isolates the ASIC status, which will indicate if the ASIC is down due to a hardware or software issue.
```

Resolution Steps

Step 1: Step: Replace the line card, Command: replace linecard BTM-ACC-23, Description: Replace the faulty line card with a functioning one to restore normal node operation.

Step 2: Step: Replace the power supply, Command: replace power supply BTM-ACC-23, Description: Replace the faulty power supply with a functioning one to restore power supply redundancy.

Step 3: Step: Replace the ASIC, Command: replace asic BTM-ACC-23, Description: Replace the faulty ASIC with a functioning one to restore normal node operation.

Post-Check Verification

```
Command: display linecard info | include BTM-ACC-23, Expected Output: Line Card Information: Card Type: Access Line Card Status: Up, Description: Verify that the line card is operational after replacement.

Command: display power supply info | include BTM-ACC-23, Expected Output: Power Supply Information: Power Supply Status: Normal Redundancy: Active, Description: Verify that the power supply status and redundancy are operational after replacement.

Command: display asic info | include BTM-ACC-23, Expected Output: ASIC Information: ASIC Type: Access ASIC Status: Up, Description: Verify that the ASIC is operational after replacement.
```

Roll-back Procedures

- Step: Restore the original line card, Command: undo replace linecard BTM-ACC-23, Description: Restore the original line card to its previous state.

- Step: Restore the original power supply, Command: undo replace power supply BTM-ACC-23, Description: Restore the original power supply to its previous state.

- Step: Restore the original ASIC, Command: undo replace asic BTM-ACC-23, Description: Restore the original ASIC to its previous state.